PERSONAL INFORMATION

Marilaure Grégoire

MAST (Modelling for Aquatic Systems) research group,
Fresh and Oceanic science Unit of research
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Full Professor at the Liège University, Directeur de recherches, Fonds National de la Recherche Scientifique.

Head of the MAST group and Director of the Fresh and Oceanic science Unit of research FOCUS

Research thematic: Environmental forecasting, Biogeochemical and physical modelling, data assimilation, marine ecosystems and biogeochemical cycles, ocean deoxygenation, traits modelling.

WORK EXPERIENCE, EDUCATION AND TRAINING

01/01/2023-	Full Professor at the Liège University
01/09/2021-31/12/2022	Professor at the Liège University
1/10/2015-31/08/2021	FNRS Research Director.
1/10/2010-30/09/2015	FNRS Senior Research Associate.
1/10/2002-30/09/2010	FNRS Research Associate.
1/12/2002-30/11/2004	Marie Curie fellowship from the EU, Netherlands Institute of Ecology (NIOZ), The Netherlands (supervisor: Karline Soetaert).
1/10/1999-30/09/2002	FNRS postdoctoral position (Chargée de recherches).
1/10/1995-30/09/1999	PhD in applied Science on "the Three-dimensional modeling of the Black Sea physics and ecosystem" obtained *with summa cum laude and congratulation of the board of examiners*) (Promotor, Jacques Nihoul, Liège University).
1/10/1995-30/06/1996	European Diploma Eruditionum Altarum in the modelling of the marine Environment (*summa cum laude*).
1/10/1990-30/06/1995	Civil engineering (in mechanics-physics), Liege University, Belgium (results obtained each year with *summa cum laude and, in 1992, 1994, 1995, congratulation of the board of examiners, with 92% in 1995*).
Career breaks	3 maternity leaves (2000, 2003, 2006) + 2 parental leaves (2003, 2006).

International EDITORIAL and REVIEWING ACTIVITIES

Since 2012, **Associate Editor** for Biogeosciences.

Guest editor of special issues in peer review journals (16)
Member of the editorial report of the Marine Copernicus Ocean State Report since 2018. From 2022, invited as chair of the Copernicus Ocean State Report published in the new Journal State of the Planet Journal, Copernicus).

International Reviewing activities

- Expert for the European Commission: evaluation of Horizon Europe, H2020 (proposal evaluation and project monitoring), ERC Consolidator, FP7, FP6, COST.
- March 2018: Member of the Evaluation Panel of the Ocean and Lake Master program handled by UA, UGent, VUB.
- Commission de sélection poste Fedtwin, « Abiotic and Biotic sediment dynamics along estuarine-marine Gradients in times of global change », Université d'Anvers (Biological Department) et le Musée des Sciences Naturelles (May-June 2022).
- Evaluation promotion Full Professor, University of Maryland, US, Février 2022.
- Reviewer for the IPCC Special Report on Ocean and Cryosphere (SROC). (June 2018)
- Expert for Research Councils and National Science Foundation/agency from UK, France, US, Switzerland, Romania, Peru,..
- Reviewer for many journals (e.g. nature communication, nature geoscience, progress in oceanography, Journal of Geophysical Research, Biogeosciences, Journal of Sea Research, Dynamic of Atmosphere and Ocean, Deep Sea Research, Journal of Marine Systems, Ocean Science,..).

PUBLICATIONS

Publication records

- Scientific seniority: **19.5 years** (as estimated from ERC starting grant rules i.e. number of years after PhD (1999) - 18 months*kid (3)). Full list on google scholar, ORBI and research gate.

INTERNATIONAL ADVISORY AND EXPERT GROUPS

Member of international advisory boards

- Since 2018, member of the Scientific and Technical Advisory Committee (STAC) of the EU Marine Copernicus Service.
- Co-chair of the advisory committee of the French MEDIATION (Methodological developments for a robust and efficient digital twin of the ocean) project (2022-2027) recently selected as Programme Prioritaire de recherches Océan & Climat launched by President Macron under the call « Un océan de solutions ».
- Member of the advisory board of the EU H2020 SEAMLESS project (2020-2023).
- Member of the scientific Committee of the UN Ocean Decade initiative « Coast Predict »
- Member of the NEMO Scientific Advisory Committee.
- Member of the Board (2018-2021) of the European Geoscience Union Fridjof medal of the European Geoscience Union medal committee.
- Since 2017, Member of the Scientific Committee of GMMC (Groupe de Mission Mercator Coriolis) part of the LEFE (Les Enveloppes Fluides et l'Environnement) program from CNRS INSU (Institut des Sciences de la Terre et de l'Univers).
- Member of the Scientific Advisory Board of the European MyOcean phase1 -project from the EU GMES program (Global Monitoring for Environmental Security).
- Member of the Steering Committee » and workpackage leader in EU projects where I am involved (e.g ; NECCTON, BRIDGE, CE2COAST,Copernicus, Sesame,...)

Member of international expert teams

- Chair of the European Marine Board Oxygen expert panel for the writing of a "Navigating to the future" policy brief ((Oxygen working group).
- 2021-2022 member of the « High-Level guidance » working group of the World Meteorological Organization for writing the guidance document towards the implementation of the WMO WIGOS vision.
- Since 2020: member of the Joint Expert Team on Earth Observing System Design and Evolution (JET-EOSDE) of the World Meteorological Organization.
- Since 2017, member of the Marine Ecosystem Analysis and Prediction Task Team, from GODAE Ocean Predict.
- 2016- Member of the Network of EU Experts for ReDeveloping Models of The European Marine Environment (DG-ENV, DG-JRC).

INTERNATIONAL NETWORK, RESEARCH PROJECTS

Chair of International network

- Co-chair (with Andreas Oschlies, GEOMAR, Germany) of the GO2NE(Global Oxygen Network) recently launched by IOC (Intergovernmental Oceanographic Commission) UNESCO.

European Projects

- PI et leader du work package « New capability in benthic habitat modelling » dans la proposition Européenne NECCTON (New Copernicus Capability for Trophic ocean networks) soumise à l'appel Horizon Europe (Horizon-CL4-2022-SPACE-01-41) (01/01/2023-31/12/2026) .
- Co-PI du projet ODESSA ("Optimal Design of Sensemble-based Simulation Systems for Applications to marine forecasting"), Appel 21036-COP-INNOSCI, Service Evolution R&D projects, (30/06/2022-29/06/2024).
- PI dans le projet européen BS-MFC, Copernicus 2 (Black sea marine forecasting centre), (01/01/2022-31/12/2024.
- PI et leader du work package "Ecosystem Dynamics under multiple stressors" du project H2020 BRIDGE (Advancing Black Sea Research and Innovation to Co-Develop Blue Growth within Resilient Ecosystems) (1/06/2021-31/12/2025).
- PI, leader du workpackage "Regional Ocean: downscaling past changes and climate projections) and Belgian representative in the JPI Ocean and Climate CE2COAST projet (01/09/2020-30/08/2023).
- Co-PI dans le projet H2020 -SPACE FORCOAST (Earth observation services for fishery, bivalves mariculture and oyster ground restoration along European Coasts (DT-SPACE-01-EO-2018-2020),
- Coordinator du projet ESA "Earth Observation for Science and Innovation in the Black Sea" (EO4SIBS, 11/06/2019-11/09/2021).
- PI dans le projet européen BS-MFC, Phase 2 (Black sea marine forecasting centre), (01/04/2018-31/12/2021.
- PI dans le projet européen BS-MFC, Phase 1 (Black sea marine forecasting centre), 2016-2018, deputy leader for the Black sea in the product quality working group, representative for the Black sea biogeochemistry for the Ocean State report.
- PI in the EU FP7 IP PERSEUS (Policy-oriented marine Environmental Research in the Southern EUropean Seas (01/2012-12/2015).
- Associate partner in the EU FP7 IP HYPOX (In situ monitoring of oxygen depletion in hypoxic ecosystems of coastal and open seas, and land-locked water bodies) (2009-2012).
- PI and Work package leader in the EU FP6 IP Sesame (Southern European Seas: Assessing and Modeling ecosystem changes).
- Marie-Curie Reintegration grant (12/2004-11/2005) on "Mathematical Modeling of Marine Ecosystems:

model development, parameterization and simplification using data assimilation".

- Scientific coordinator of the EU ARAL-KUM project (INCO) devoted to the study of the desertification process in the Aral Sea basin
- Coordinator of the EU INTAS project (Study of the nitrogen Cycling in the Oxic / Anoxic Environment of the Black Sea) (01/06/00-31/11/02)
- Scientific coordinator of the EU VENTIL project (INCO) devoted to the study of the ventilation process in the Black Sea (10/97-10/00)

National/regional Projects

- Coordinator of the FullContinuum project (FNRS PDR, 1/01/2021-31/12/2024) on seamless coupling of the ocean and river system.
- Coordinator of the BENTHOX project (FNRS PDR, 1/10/2015-30/09/2019) on the understanding of hypoxia in the Black sea Budget.
- PI in the Face-it project (BRAIN call, Belspo) on the (Functional biodiversity in a Changing sedimentary Environment: Implications for biogeochemistry and food webs in a managerial setting, the North Sea) (2015-2019).
- Involved in the Belgian Science Policy project " PEACE" on the "role of PElagic cAlcification and export of CarbonatE production in climate change", (2005-2009).
- PI of an "Action de recherches concertées » from the French speaking community, " RACE : Rapid Assessment of the Marine Coastal Environment"» (2005-2010) .
- Co-Principal Investigator of "Mandat d'impulsion scientifique" from FNRS, " Understanding the carbon cycle in the coastal zone". (2005-2008).

INTERNATIONAL CONFERENCES, WORKSHOPS

Invited and Keynote talks (invited seminars not mentioned)

- Marine research at the Liège University: a long tradition of oceanography far away from the sea, VLIZ young scientific day, Bruges, 25 February 2005.
- A recipe for biogeochemical modelling using 3D pelagic models, Istanbul, EU HYPOX meeting, March 2010.
- The assessment of the past decades based on modelling and data analysis: some examples from the EU Sesame project. Venice, CIESM meeting, May 2010.
- Integrated tools for environmental assessment in the Black Sea, Umbrella meeting of the European PERSEUS project.
- The challenging management of the good environmental status (GES) of marine waters: the Black Sea case, NIOZ Science days (Invited Keynote, the Netherlands, June 4th 2013).
- Keynote Speaker at the 8th International Workshop on Modeling the Ocean, Bologna, June 7-10 2016 (invited by the organizing committee).
- Invited speaker at the 18th Annual Large Marine Ecosystems and Coastal Partners Meeting, IOC/UNESCO Headquarters, Paris (France) 6 - 8 December 2016. (invited by IOC Unesco)
- Invited webinar, Deoxygenation in the global and coastal ocean: challenges of observing and modelling low oxygen zones, June 2018 (invited by the Global Ocean Observation System , GOOS)
- Plenary speaker at the Romanian Parliament for giving a talk on the Black Sea : Knowledge to promote recovering at an international event (see here) organized in the frame of the Romanian Precedence of the EU. (Invited by the organizers of the event, Adrian Stanica).
- Keynote Speaker at the 3rd Baltic Earth Conference, Earth system changes and Baltic Sea, Jastarnia, Het Peninsula, Poland, June 1-5th 2020. (online).(Invited by the organizing committee)
- Invited speaker at the 2nd International Conference on Oceanography for West Asia (RCOWA-2020), 16-17 September 2020, Republic of Iran.(Invited by the organizing committee)

- Invited Speaker at the 50th Birthday of the NIRMD and 50th Black Sea day, October 2020.
- Invited as a Discussion leader at **Gordon Research Conference ("Marine Biogeochemistry"), Barcelone, May 1-5, 2022.**
- Invited as a speaker at the excellence network MEOPAR network November 28th 2022. State of Ocean Deoxygenation in Canada, Toward a community of practice … or sort of..?

Organisation of international conferences, sessions, workshops

- Organizer of the International Liege colloquium on ocean dynamics in 2001, 2011, 2013, 2014, 2021, 2022 (~200 participants).
- Member of the Scientific Organizing Committee of the GO2NE Webinars on ocean deoxygenation (monthly events).
- Member of the Scientific Organizing Committee of the World Meteorological Organization Scoping workshop on Earth System Modelling, Geneva, December 9-11th, 2019 (discussion leader for the ocean session).
- 2019: co-organization of the oxygen data platform scoping workshop, Sopot, Poland, November 11-12th, 2019.
- Member of the Scientific committee of the workshop on Towards the Future of Marine Computation for Sustainable Development, June 2019.
- Member of the scientific committee and session co-chair at the Ocean deoxygenation conference, Kiel, Germany, September 2018.
- 2014: Member of the Scientific Committee of the 8th European Conference in Ecological Modelling, Marrakech, 27-30 October 2014.
- 2015: member of the organizing committee of the Baltex-Hymex workshop (Rome Nov 4-6, 2015)
- 2011-2016: (Co)-Convener of a session on "Advances in understanding of the multi-disciplinary dynamics of the Southern European Seas (Mediterranean and Black Sea)" and in 2013-2015 Coastal ecosystem under anthropogenic pressure: impact on ecosystem structure and services at the EGU (European Geophysical Union) meeting.
- Co-Chair with R. Feely of a session of Low PH and O2 environments during the 2nd International Ocean Research Conference organized by UNESCO/IOC and the Oceanography Society (TOS) .
- Co-chair with V. Martinez Vincente of the session on ocean health during the ESA Ocean Science Cluster Collocation meeting (November 2021)

Panelist at international events

- 2014: 17-21 November, Barcelona: Co-Chair with R. Feely of a session of Low PH and O2 environments during the 2nd International Ocean Research Conference organized by UNESCO/IOC and the Oceanography Society (TOS).
- Panelist at the round table of the final meeting of the EU PERSEUS project on "Where do we go from here in the SES?"(December 9th, 2015)
- Panelist at the workshop on the use of marine Copernicus data for assessing the Marine Strategy Framework Directive, Oct 2nd, 2019 (Invited by Mercator Ocean International).
- Panelist at the 7th European Marine Board Form on the use of Bug data in marine Science, European Green Deal, The EU Biodiversity strategy, and Digital Twin of the Ocean, October 23rd, 2020, Invited by the Eu Marine Board).

PhDs students, committees, juries

Promoter of PhD thesis

- Caroline Raick, 2006. Physical Engineer, Thesis in Applied Sciences, Mathematical modeling of the Ligurian Sea ecosystem using models of different complexities. Application of a Kalman filter to improve models reliability, 2006, 278p.
- Pascal Joassin, 2011. Mathematical modeling of biogeochemical processes associated to a bloom. Study of the seasonal and long-term variability of biogeochemical properties in the Black Sea using a Data Interpolating Variational Analysis (DIVA), 250p.
- Arthur Capet, 2014. Study of the multi-decadal evolution of the Black Sea hydrodynamics and biogeochemistry using mathematical modelling, 256p.
- Evgeny Ivanov, (foreseen for 2021) ongoing: Modelling the hydrodynamic of the North Sea in a managerial setting.
- Florian Ricour, ongoing, Development of an oxygen climatology for the world ocean.
- Catherine Meulders: Modelling Black Sea's ecosystem functioning and services in a climate change context.
- Lauranne Alaerts: Seamless modelling of the Danube-Delta coastal Black sea.

Rapporteur

- Frédéric Diaz, 2014 Aix-Marseilles Université).« Thèse d'habilitation à diriger des recherches (France)
- Yunchang He, 2012 (University of Oldenburg, Germany),
- Fayçal Kessouri, 2015. Université de Toulouse.
- Florent Garnier, 2016.. Université Grenoble Alpes.
- Nicolas Mayot, 2016. Université Pierre et Marie Curie.
- Bachelery Marie-Lou, 2016. Université de Toulouse.
- Camille Richon, 2017. Université Paris-Saclay.
- Giulia Mussap, 2017., University of Cadiz (President of the Jury).
- Damien Couespel, 2018 Sorbonne Université
- Marine Bretagnon, 2018. Université de Toulouse.
- Joëlle Habib, 2023, Université de Toulouse

External reviewer for PhD or member of the Jury

- Margarita María Rincón Hidalgo, (01/2015) (Universidad de Granada, Spain).
- Ali Aydogdou, 2016. Universitry of Bologna.
- Alexandra Steckbauer, 2016. University of Barcelona.
- Rémi Pages, 2020, Aix-Marseille Université

TEACHING

Teaching at ULiege (2019-)

- INFO0956: Introduction to biological data analysis, 5h Th, 20h TD, Grégoire Marilaure, MEYER Patrick.
- BIOL0030-1 : Modeling dynamical biological systems, 15h Th, 15h TD, Grégoire Marilaure, MEYER Patrick.
- OCEA0014-1:Méthodes mathématiques d'analyse et de modélisation appliquées à l'environnement, 20h Th, 20h Pr, GRÉGOIRE Marilaure
- OCEA0057-7: Marine Ecology. Part: Introduction to Marine Ecosystem Modelling, 15h Th., 15 Pr. Grégoire Marilaure
- OCEA0089-1: Introduction to Marine Ecosystem Modelling, 15h Th., 15 Pr. Grégoire Marilaure
- OCEA0090-1: Dynamique des écosystèmes marins, 20h Th, 20h Pr. Grégoire Marilaure
- OCEA0096-1 : Modélisation des écosystèmes et des cycles biogéochimiques, 15h T., 30h Pr. Capet Arthur, Grégoire Marilaure Munhoven Guy.
- OCEA0224-1 : Analyse statistique des données océanographiques, 15h Th, 15h Pr, CAPET Arthur, GRÉGOIRE Marilaure, MEYER Patrick

Teaching at international

- Co-Director of the IOC-Unesco GO2NE Summer School (Xiamen University, September 2019).
- Teaching at the Face-IT summer school on the integrated study of hard and soft sediment using

summer school

field work and modelling. (July 2017, October 2019)

- Teaching at the International Sesame Summer school on Ecological modelling (Malta, June, 2009).

AWARDS, DISTINCTIONS

- Laureate of the Scientific Award delivered by "L'Association des Ingénieurs de Liège (AILg)" (best PhD in Applied Sciences). (2002)
- Laureate of the André Leroux award (best PhD in Applied Sciences) delivered by "Les Amis de L'Université de Liège".(2002)
- Marie Curie Individual International Fellowship of the European Community.(2002)
- CGRI travel ship to stay to weeks at the University of Grenoble (LEGI), France. (2002)
- Laureate of the American Geophysical Union Student Travel Grant program. (1998)
- Laureate of the IRMA (Jeunes océanographes) award of the best PhD student presentation.(1998)

COMMUNITY SERVICE

Responsabilities at ULiege

- Since February 2019, elected as Director of the Freshwater and OCeanic science Unit of reSearch (FOCUS) research unit
- Since 2020 Member of the University Research Council (Science and Technics sector).
- Since January 2019, member of the FNRS council of administrators
- President of the Doctoral College in Oceanography at the Liege University.
- Since 2016, Member of the Research and Teaching-Research commissions of the Science Faculty.
- Member of the council of studies in oceanography
- Participation in round table for PhD formation organized by the University

Science Diffusion

- Coordinator of a lectures series for the Collège de Belgique on "L'Océan à travers les temps : mécanismes d'une épopée passionnante perturbée par l'Homme ». Intervenants : Anne-Christine Da Silva, Xavier Fettweis, François Primeau, François Gemenne et Philippe Vincent.
- Novembre 2019, Intervenant dans le cadre du colloque de la société des Sciences de Liège, « Le changement climatique :des outils diagnostiques aux solutions technologiques ».
- 6 Août 2019 : Intervention dans Jour Première, Sophie Léonard, huit mesures à adopter de toute urgence pour préserver le rôle des océans dans le maintien de la vie sur Terre, commentaire des conclusions du rapport IPSO.
- May 2019: Université du temps libre Paris,
- May 2019: U3, Liège
- 12 Juin 2018, organisation d'un café des Sciences, le Grand bleu et l'Homme.
- June 2018 : Publication d'une fiche technique de 15 pages à destination des décideurs et du grand public sur la désoxygénation des Océans, IOC-UNESCO, Paris. (Breitburg, Grégoire and Isensee, and the GO2NE group
- March 2018: Printemps des Sciences
- March 13th 2018: participation to the France Culture broadcast "la Méthode Scientifique" animated by Nicolas Martin on "Ocean chemistry modifications".
- Janvier 2018, interview on the deoxygenation process in the Ocean in relation to the Science paper, 5/01, Soir Première, Nicolas Vandenschrick ; 4/01 le Soir, 12/01 Journal de 19h00 RTL TVI, 6/01 Janvier 2018 : la Meuse).
- June 2018 : Publication of a technical brief for decision makers, IOC-UNESCO
- June 2017: UN Ocean conference New York, Organization with the GO2NE group of a side event on Ocean deoxygenation.